

than down, which would result in capital losses for the bonds held in the funds.

As we have discussed, changing interest rates lead, in turn, to capital gains or losses for investors. For professional bond traders, rising interest rates would be a serious problem for someone who had just purchased a long-term bond. Most traders have no intention of holding bonds to maturity and will realize capital losses on the subsequent sale. Interest rate increases might also force retirees owning bond funds into a position of selling shares at a loss in order to meet retirement expenses. For a bond portfolio that is not fully immunized, this triggering of sequence-of-returns risk can create irreparable harm for retirees.

I would argue that it is much easier for a retiree to ignore unrealized capital losses on an individual bond than for a professional trader or retiree needing to sell bond shares to meet expenses, because the individual bond is bought with the purpose of being held to maturity to provide a desired amount of income at that date.

Exhibit 7.9 quantifies this point. If interest rates are at 2 percent when a thirty-year bond is purchased, but then subsequently rise permanently from 2 percent to 3 percent during the first year the bond is held, the retiree is sitting on an unrealized capital loss for the next twenty-nine years. After one year, the bond price falls to \$816, representing an 18.4 percent loss. Nevertheless, the bond continues to pay its 2 percent coupon payments and at maturity will repay the \$1,000 face value. As the maturity date slowly approaches, the unrealized losses slowly dissipate. The bond price gradually returns to match its face value. The full recovery will happen at the maturity date when the final cash flows are received as expected. In this way, a household investor can be justified in ignoring those unrealized capital losses on the bond, as they will not be realized if the bond is held to maturity.

Exhibit 7.9

30-Year Maturity Bond with 2% Coupons and \$1,000 Face Value

Ongoing Price of Bond if Interest Rates Stay at 2% and if Interest Rates Rise to and Stay at 3%